JUAN SEBASTIÁN SÁNCHEZ URBANO

Junior Data Engineer · ETL Automation, Cloud (AWS), Machine Learning & AI

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LinkedIn · GitHub · Portfolio

PROFESSIONAL PROFILE

Industrial Automation Engineer and Master's student in Data Science and Artificial Intelligence, with a Junior Data Engineer profile. I design and build ETL pipelines with medallion architecture (Bronze/Silver/Gold), work with relational databases (SQL, MySQL), program in Python and R, and develop automation workflows with n8n, REST APIs and AI agents. I combine industrial engineering with data analytics to ensure the integrity, availability and quality of information, under continuous improvement methodologies (PDCA). I maintain a public portfolio with projects, my own tools and data, automation and 3D printing services.

TECHNICAL SKILLS

Data Engineering & Cloud: ETL pipelines, Medallion architecture, AWS, REST APIs, Advanced SQL, MySQL, Data modeling and consolidation, Data quality and cleaning.

Languages & Data Science: Python, R, SQL, Machine Learning, Predictive modeling, Classification, Statistical analysis, EDA, Git and GitHub.

Automation, BI & IoT: n8n and AI agents, AI chatbots, Power BI, Advanced Excel, HTML dashboards, Real-time IoT sensors and instrumentation.

Web Development & Deployment: HTML, CSS, JavaScript, Bash, YAML, GitHub Actions, GitHub Pages.

Industrial Engineering: Industrial automation, PLC programming, CNC and robotics.

Methodologies & Management: PDCA (continuous improvement), Information validation and integrity, Analytical problem solving, Results orientation, Effective communication.

PROFESSIONAL EXPERIENCE

Technical, Sales and Data Automation Advisor — Proeléctricos

2019 – Nov 2025 · Mar 2026 – Present

- I analyze and validate customers' technical and operational information to structure comprehensive solutions, ensuring the quality and traceability of commercial records.
- **Achievement:** I designed and implemented an AI chatbot for the web channel, integrated into an n8n automation pipeline that automatically extracts, validates and routes reports and information to the responsible areas.

CNC and Robotics Area Lead (Production Data Analysis) — Metálicas Mundial

Oct 2025 – Jan 2026

- I led the capture and analysis of critical production and quality-control variables, ensuring the reliability of manufacturing process data.
- **Achievement:** I identified bottlenecks and optimized cycle times by analyzing machinery operating data under agile continuous-improvement methodologies, increasing the area's productivity.

FEATURED PROJECTS

End-to-end automated ETL pipeline

MinTIC / Talento Tech · 2026 · Python, n8n, REST APIs, ETL

- An extraction, transformation and loading architecture with Python, n8n and REST API calls to consolidate centralized databases.
- **Impact:** 100% of manual data entry eliminated, with complete, standardized and reliable data for executive analysis.

SIVIGILA epidemiological ETL 2018–2022

Dengue · SARI · Tuberculosis · Python, Excel, Medallion, HTML dashboard

- An ETL process over five National Institute of Health Excel files, with extraction, transformation and loading scripts, medallion architecture and an HTML dashboard to visualize results.

Kaismart Solutions ETL lab

Universidad Autónoma de Occidente · MySQL, Python, Excel, Medallion, Schedule

- Integration of a MySQL sales table (5,000 records) with 50,000 logistics events in Excel, joined by order ID. Includes exploratory analysis, medallion architecture, 10 business questions and scheduled execution.

Web portfolio with medallion architecture

2026 · HTML, CSS, JavaScript, Python, GitHub Actions

- Site published on GitHub Pages with a repository in Bronze/Silver/Gold layers, an automated Python build and continuous deployment with GitHub Actions. Includes my own tools: an HTML reader with safe preview, a WhatsApp link generator and a JSON formatter.

Real-time data capture and alerts (industrial IoT)

Engineering projects · 2022 – 2024 · IoT, Sensors, Alerts, Real time

- Integration of industrial sensors and instrumentation for real-time data ingestion, automatic validation of critical operating variables and timely control alerts.

3D Workshop and automation services

Business venture · 3D printing, n8n, AI, WhatsApp

- Custom products (printed keychains and cases, 3D printing) and automation services with n8n, AI content and industrial engineering, with a product designer connected to WhatsApp.

EDUCATION

Master's in Data Science and Artificial Intelligence · Universidad Autónoma de Occidente

In progress

- Cloud architecture and services (AWS), Machine Learning models, ETL process design, statistical programming in R and Python, and advanced analytics.

Industrial Automation Engineering

Graduate engineer